

Steph Buongiorno, PhD

Curriculum Vitae

Department of Data and Decision Sciences
Emory University
Interdisciplinary Sciences Building
36 Eagle Row, Atlanta GA 30322

sbuongi@emory.edu

Education

- | | |
|------|---|
| 2023 | PhD, Applied Engineering, Southern Methodist University <ul style="list-style-type: none">• Subject areas: Computational History, Video Games, Computer Science |
| 2017 | MA, English, West Virginia University |
| 2014 | BA, English, University of Texas at Arlington (Summa Cum Laude) <ul style="list-style-type: none">• Minor: Linguistics |

Employment: Research Positions

- | | |
|--------------|---|
| 2025-present | Postdoctoral Fellow, Faculty, Department of Data and Decision Sciences, Emory University |
| 2025-2027 | (Declined) Open Science Postdoctoral Fellow, Faculty, Libraries, Carnegie Mellon University |
| 2023-25 | Postdoctoral Research Fellow, Guildhall Video Game Department, Southern Methodist University |
| 2023-25 | Democracy Lab Co-Director, Emory University
Topic: Research, publications, and development of an advanced text mining, data sharing, and data visualization public web app for the digital humanities (with 3-8 undergraduate students at a given time). |
| 2022-23 | Technical Lead, "The Human Trafficking Project," Department of Computer Science, National Institute of Justice Grant (H.R. 2471), PI Corey Clark, Southern Methodist University |
| 2022-2023 | (Declined) Laboratory for Social Minds, Dietrich College, Carnegie Mellon University |
| 2021-22 | Research Assistant, "Toward a History of the Associative-Developmental State," Department of History, Southern Methodist University |

2021-22	Research Assistant, “The History of Modern China,” Department of History, Southern Methodist University
2018-21	Technical Lead and Project Manager, “Global Urbanization and Housing Affordability: Poverty, Property, and the City,” National Science Foundation Grant (no. 1520103), PI Jo Guldi, Department of History, Southern Methodist University

Employment: Teaching Positions

2025-present	Postdoctoral Fellow, Faculty, Department of Quantitative Theory and Methods, Emory University
2025-2027	(Declined) Open Science Postdoctoral Fellow, Faculty, Libraries, Carnegie Mellon University
2023-25	Postdoctoral Fellow, Guildhall Video Game Department, Southern Methodist University Topic: Research and writing for theses (for master’s students across four specializations: programming, level design, art, and production).
2023-25	Democracy Lab Co-Director, Emory College Topic: Research, publications, and development of an advanced text mining, data sharing, and data visualization public web app for the digital humanities (with 3-8 undergraduate students at a given time).
2025	Part-Time Lecturer, iSchool, University of Texas Topic: Introductory Python programming for informatics students.
2023	Support Instructor, Team Game Production I, Guildhall, Southern Methodist University Topic: Reproducibility and version control.
2022-23	Teaching Fellow, Foundations and Applications of Humanities Analytics, National Endowment of the Humanities Grant (no. HT-272418-20), Santa Fe Institute Topic: Digital humanities and cultural analytics.
2021-22	Support Instructor, Topics in Digital History, Southern Methodist University Topic: Code, bias, and the representation and obfuscation of gender in both coding libraries and textual corpora (e.g. U.S. Congress and Reddit).
2017-18	Graduate Teaching Assistant, Academic Writing, Southern Methodist University Topic: Introduction to academic research and writing.
2017	Assistant SCUBA Instructor, Department of Physical Fitness, Southern Methodist University.

2015-17	Graduate Teaching Assistant, Composition and Rhetoric, West Virginia University Topic: Introduction to academic research and writing.
2010-12	Assistant SCUBA Instructor, Department of Exercise and Sport Activity, The University of Texas at Arlington
2010-12	Assistant SCUBA Instructor, Texas Women's University

Employment: Related Positions

2023-present	Instructor, Guildhall Academy (video game design for high school students), Guildhall, Southern Methodist University
2022-23	Computer Science Senior Design, Southern Methodist University

Awards

2023	Dean's Award for Best Poster Presentation in Computer Science, Southern Methodist University
------	--

Publications

Peer Reviewed Books

Steph Buongiorno [**Corresponding Author**] and Jo Guldi. *Text Mining for Historical Analysis*. Under contract at Cambridge University Press. Forthcoming June 2025.

Steph Buongiorno [**Corresponding Author**] and Jo Guldi. *Text Mining for Historical Analysis*. eBook edition. Under contract at Cambridge University Press. Forthcoming June 2025.

Peer Reviewed Articles

Jo Guldi, Peter Kastor, and Steph Buongiorno [**Last Author**]. "Word Embeddings as a Key to the Study of Bias, Race, and Gender in Congress, 1880-2010." *Journal of Digital History*. **Forthcoming**.

Peer Reviewed Conference Proceedings

Sangmitra Madhusudan, Trush Shashank More, Steph Buongiorno [**Third Author**], Renata Dividino, Jad Kabbara, Ali Emami. "Common to Whom? Regional Cultural Commonsense and LLM Bias in India". Association for Computational Linguistics (ACL), San Diego, CA, USA, 2026.

Steph Buongiorno [**Corresponding Author**], Jake Klinkert, Tanishq Chawla, Zixin Zhuang, and Corey Clark. "PANGeA: Procedural Artificial Narrative Using Generative AI for Turn-Based Video Games." *Proceedings of the 2024 Association for the Advancement of Artificial Intelligence (AAAI) Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, Lexington, KY, USA, 2024.

Jake Klinkert, Steph Buongiorno [**Second Author**], Corey Clark. "Evaluating the Efficacy of LLMs to Emulate Realistic Human Personalities." *Proceedings of the 2024 Association for the Advancement of Artificial Intelligence (AAAI) Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, Lexington, KY, USA, 2024.

Steph Buongiorno [**Corresponding Author**], and Corey Clark. "Leveraging Gaming to Enhance Knowledge Graphs for Explainable Generative AI Applications." *Proceedings of the 2024 Institute of Electrical and Electronics Engineers (IEEE) Conference on Games (CoG)*, Milan, Italy, 2024.

Peer Reviewed Essays

(By invitation) Steph Buongiorno [**Corresponding Author**]. "Radically Slow Pedagogy in the Age of AI." *Early American Studies*, 2026. **Forthcoming**.

Published Abstracts

Kiril Avramov [**Equal Contribution**], Steph Buongiorno [**Equal Contribution**], Adam Hanzel, and Thorsten Ries. “Who is the Working Class, Socialist James Bond? Exploring Propaganda through Knowledge Graphs and AI-Assisted ‘Forensic Reading’ in Bulgarian Socialist Spy Fiction.” *International Association for Intelligence Education Annual Global Conference*.

Articles Under Review

Steph Buongiorno [**Corresponding Author**], Jo Guldi [**Corresponding Author**], Marnie Hughs-Warrington [**Corresponding Author**], Gregory J. Palermo, Sohan Bellam, Rosie Larson, and Samuel Jiang. “Multi-Scale computational analysis of Wikipedia’s telling of global history.” *Nature Humanities and Social Sciences*.

Steph Buongiorno [**Corresponding Author**], Alexander Cerpa, Jo Guldi. “Hansard 2.0: Discovering Lost Records and the Creation of an Analysis Ready Data Set.” *Journal of Cultural Analytics*. Revise and resubmit.

Kiril Avramov [**Equal Contribution**], Steph Buongiorno [**Equal Contribution**], Adam Hanzel, and Thorsten Ries. “Who is the Working Class, Socialist James Bond? Exploring Propaganda through Knowledge Graphs and AI-Assisted ‘Forensic Reading’ in Bulgarian Socialist Spy Fiction.” *Journal of Intelligence History*. Recently submitted.

Papers In Development

Steph Buongiorno [**Corresponding Author**], Serena Di Martino, and Ryan Schaefer. “Democracy in Data: An Open Infrastructure for Text Mining Current and Historic Data.” Target: *Historical Methods: A Journal of Quantitative and Interdisciplinary History*.

Steph Buongiorno [**Corresponding Author**] and Adriana Chira. “Mapping the Elusive: Using Network Analysis to Understand Slavery, Debt Relations, and the Emergence of a Free Population of African Descent in 18th-Century Colonial Havana.” Target: *Journal of Digital History*. Estimated submission: Fall 2025.

Laurie Giddens, Stacie Petters, and Steph Buongiorno [**Last Author**]. “Vigilant Interactions and Information Sharing within Public-Private Partnerships.” Target: *Journal of Management Information Systems on Critical National Infrastructure*.

Steph Buongiorno [**Corresponding Author**], Rosie Larson, Meena Ramswamy, Greg Palermo, and Jo Guldi. “Tracing a Distinctive Feminized Democratic Lexicon, 1917-2025: New Data and Methods for Analyzing American Political Development.”

Adam Hanzel, Thorsten Ries, Hiba Faruqi, Kiril Avramov, Steph Buongiorno [**Last Author**]. “Alternative AI and Digital Authoritarianism: Analyzing AI as a Cultural Object.”

Steph Buongiorno [**Corresponding Author**]. “Beyond the Black Box: Toward Explainable AI for Computational Text Analysis in the Digital Humanities.”

Other Contributions

Datasets

Nan Jiang, Rosie Larson, Steph Buongiorno, and Jo Guldi, 2025, "Replication Data for: Multi-Scale Computational Analysis of Wikipedia's Telling of Global History," <https://doi.org/10.5281/zenodo.17527577>, Zenodo.

Rosie Larson and Steph Buongiorno. "The United States Congressional Record (1873-Present): Parsed Speeches, Debate Titles, and Identified Speakers",

Adriana Chira; Aisnara Perera Diaz; Maria de los Angeles Merino Fuentes; Steph Buongiorno, 2025, "Escrituras Protocolos 1640 al 44 y 1650", <https://doi.org/10.7910/DVN/AEDRPT>, Harvard Dataverse, DRAFT VERSION

Adriana Chira; Aisnara Perera Diaz; Maria de los Angeles Merino Fuentes; Steph Buongiorno, 2025, "Escrituras Protocolos 1730 to 1733", <https://doi.org/10.7910/DVN/AZCY4B>, Harvard Dataverse, DRAFT VERSION, UNF:6:gj/M2efvaPDnYElSfLR5A== [fileUNF]

Steph Buongiorno; Robert Kalescky; Omar Alexander Cerpa; Jo Guldi, 2022, "The Hansard 19th-Century British Parliamentary Debates with Improved Speaker Names: Parsed Debates, N-Gram Counts, Special Vocabulary, Collocates, and Topics", <https://doi.org/10.7910/DVN/ZCYJH8>, Harvard Dataverse.

Steph Buongiorno; Omar Alexander Cerpa; Jo Guldi, 2022, "The Hansard 19th-Century British Parliamentary Debates with Improved Speaker Names: Speaker Metadata", <https://doi.org/10.7910/DVN/Z3LTVV>, Harvard Dataverse, FORTHCOMING

Open Source Software Development

Steph Buongiorno, Serena Di Martino, Ryan Schaefer, Shawn Chen, Alisha Morejon, Hailey DeMark, Francesca La Marca, Grace Schultz, Wes Anderson, Chris Miller, and Matt Swigart. [Democracy Viewer](#), Emory College. *I lead the on-going development of a public-facing web app for exploring, text mining, and visualizing humanities and social sciences data sets in languages including (but not limited to) English, German, Spanish, and French. Recently adopted by Emory's Center for AI Learning.*

Steph Buongiorno. [The Hansard Viewer](#). Southern Methodist University, 2022. A Shiny app for text mining and visualizing the 19th-century British parliamentary debate using data science metrics.

Steph Buongiorno. [The Congress Viewer](#). Southern Methodist University, 2022. Description: A Shiny app for text mining and visualizing the U.S. Congressional Records using data science metrics.

Steph Buongiorno. [nasa](#). ROpenGov, CRAN. Description: An R package for accessing and downloading data from NASA APIs, including Mars rover photos, Astronomy Picture of the Day, Earth imagery, and natural events. ~400 downloads / month.

Steph Buongiorno. [noaa](#). ROpenGov, CRAN. Description: An R package for querying a clean version of NOAA climate and weather data. ~200 downloads / month.

Steph Buongiorno. [usdoj](#). ROpenGov, CRAN. Description: An R package for creating a structured version of the U.S. Department of Justice press releases, blogs, and records. ~800 downloads / month.

Steph Buongiorno. [uscongress](#). ROpenGov, CRAN. Description: An R package for creating a structured version of the U.S. Congressional Records. Parses raw text into delimited fields for record URL, speaker, title, and body text. ~250 downloads / month.

Steph Buongiorno. [oldbailey](#). ROpenGov, CRAN. Description: An R package for creating a structured version of the Old Bailey criminal trials. Handles broken tags and messy data and returns an analysis-ready dataset.

Steph Buongiorno. [hansardr](#). GitHub. Description: An R package for querying a clean version of the 19th-century Hansard Corpus.

Steph Buongiorno and Omar Alexander Cerpa. [hansard-speakers](#). GitHub. Description: Code for disambiguating speakers in the 19th-century Hansard Corpus using Levenshtein distances and parallel computing.

Steph Buongiorno and Omar Alexander Cerpa. [posextract](#). GitHub. Description: A Python package for extracting grammatical subject-predicate triples from data. Tailored for the analysis of agency in text.

Steph Buongiorno. [posextractr](#). GitHub. Description: An R package for extracting grammatical subject-predicate triples from data. Tailored for the analysis of agency in text.

Steph Buongiorno and Jo Guldi. [democracy-lab](#). GitHub. Description: A code repository for text mining techniques for the Digital Humanities.

Ryan Schaefer and Steph Buongiorno. [dhmeasures](#). CRAN. Description: Optimized, “white-box” statistical functions for textual analysis.

Steph Buongiorno. [tmha.data](#). GitHub. Description: Data for *Text Mining for Historical Analysis*.

Steph Buongiorno. “[Foundations and Applications of Humanities Analytics 2023](#).” Santa Fe Institute. GitHub. *I was the lead instructor for the SFI’s “Humanities Analytics” summer camp two years running. My code – shared here as a Github page – formed the basis for all instruction and activity.*

Steph Buongiorno. “[Foundations and Applications of Humanities Analytics 2022](#).” Santa Fe Institute. GitHub.

Steph Buongiorno. “[Digital History](#).” Southern Methodist University. GitHub.

Zixin Zhuang, Pravin Ramona, Corey Clark, Steph Buongiorno. “[PANGeA](#).” Southern Methodist University. Gitlab.

Video Game Development

Dark Shadows

Steph Buongiorno, Jake Klinkert, Tanishq Chawla, Zixin Zhaung, and Corey Clark. *Dark Shadows*, Southern Methodist University, Guildhall. Topic: Address the real-world problem of human trafficking while playing a video game. I conceived of Dark Shadows as a film noir-style “document thriller” (inspired by *Papers, Please!* and *Night Call*). I designed and developed underlying “human-in-the-loop” mechanics to collect player feedback and train a machine learning model, which can disambiguate speakers, locations, and events in real human trafficking data. I guided graduate students on the design of the gameplay mechanics and narrative.

<https://www.cbsnews.com/texas/news/i-team-video-game-helps-researchers-fight-sex-trafficking/>. I

designed NPCs that leverage large language models (LLMs) and a self-reflection-based validation system to enable dynamic, free-form interactions with the player aligned with a procedural game narrative (re: PANGeA). NPCs express traits from the Big 5 Personality Model in their responses, while the validation system ensures responses stay on narrative track. <https://www.youtube.com/watch?v=ys7vTkUZ8ic>

Other Contributions

20 visualizations in Jo Guldi’s *The Dangerous Art of Text Mining*.

Acknowledgment for support with extracting data for “Determining economic factors for sex trafficking in the United States using count time series regression” by Jang, Y., Sundararajan, R.R., Barreto-Souza, W. et al. *Empirical Economics* 67, 337–354 (2024). <https://doi.org/10.1007/s00181-023-02549-w>

Grants and Awards

Funded:

Title: Access to GADI NSI Super Computer

Award: \$5,000

PI: Marnie Hughes-Warrington

Summary of Work:

For use of National Computational Infrastructure's (NCI) supercomputer, Gadi. I am translating over 250,000 Wikipedia articles using AI to derive new insights into worldviews embedded in different articles and to create a dataset for telling a new form of history, global history.

Title: Human Trafficking Project

Opportunity: National Institute of Justice (no. 15PNIJ-22-GK-00246-BRND)

Award: \$1,187,000

PIs: Beth Wheaton, Corey Clark, Raanju R. Sundararajan

Summary of Work:

I investigated and published on how we can use knowledge graphs to encode domain information and guide generative responses, and how we can source this knowledge through crowdsourcing responses from video game players. This approach could be used to guide responses by LLMs, grounding them in community-sourced knowledge and fact. While this approach was used to improve the use of AI for the domain of human trafficking—where grounding responses in fact and avoiding hallucinations is key—this approach could also be used for meaning-making in digital humanities contexts. In pursuing this research I created open source software: Dark Shadows (the video game), the usdoj R package,

Title: Foundations and Applications of Cultural Analytics in the Humanities

Opportunity: National Endowment of the Humanities (no. FAIN: HT-272418-20)

Award: \$247,932

PIs: David Kinney and Simon DeDeo

Summary of Work: At the Santa Fe Institute's Foundations and Applications of Humanities Analytics program, I contributed to the broader adoption of computational text analysis by the digital humanities by developing instructional Jupyter notebooks and leading demonstrations of state-of-the-art methods. My work introduced participants to advanced techniques in text mining and natural language processing, emphasizing their applications within interdisciplinary humanities research.

Title: Global Urbanization and Housing Affordability: Poverty, Property and the City

Opportunity: National Science Foundation Grant (no. 1520103)

Award: \$1,000,000

PIs: Jo Guldi, Matthew Desmond, Frank Zephyr

Summary of Work: My research investigated how text mining can be applied to analyze historical change over time, with a particular focus on developing methods that support what I term “computational historical thinking.” This line of inquiry has culminated in my forthcoming book, *Text Mining for Historical Analysis*, under contract with Cambridge University Press. As part of this work, I have developed multiple open-source software applications designed to facilitate the computational analysis of historical texts (e.g.

Hansard Viewer, Congress Viewer, the oldbailey R package, and the usdoj R package), including the conceptual framework and technical blueprint for Democracy Viewer, which is now a public-facing app hosted by Emory College.

Rejected

Title:

Opportunity: Wikimedia Foundation

Award: \$5,000

PI: Jo Guldi, Steph Buongiorno, and Greg Palermo

Summary of Work:

Title: Dissentometer: Humanities-Grounded Tools for Mapping Patterns of Dissent and Consensus

Opportunity: Mellon Foundation

Award: \$5,000

PI: Jo Guldi, Steph Buongiorno, and Greg Palermo

Summary of Work: Dissentometer is a humanities-grounded analytical toolkit designed to make patterns of dissent, disagreement, and minority reasoning legible across historical and contemporary textual corpora. Drawing on methods from history, political theory, and data science, the project develops transparent metrics and visualizations that distinguish evidence, judgment, and consensus formation rather than collapsing complexity into predictive scores. By foregrounding dissent as a constructive and historically patterned force, Dissentometer offers an alternative to dominant AI systems that normalize agreement and erase conflict.

Title: “Explainable AI in Digital Humanities Research: Knowledge Graphs and SHAP”

Submitted: National Endowment of the Humanities, Dangers and Opportunities of Technology: Perspectives from the Humanities

Size: 2 years, \$75k

PIs: Thorsten Ries (Comparative Literature and Germanic Studies) and Steph Buongiorno (Guildhall)

Summary:

This Digital Humanities (DH) project focuses on making AI “black boxes” in ML-/AI-based DH applications readable utilizing Explainable AI (XAI) methods (Knowledge Graphs, KG; Shapley Values, SV, SHAP), in order to enable discipline-wide adoption of the concept and technology in “Critical AI” studies and DH. KG and SV are active research areas in CS with the goal to trace and mitigate issues with reliability, robustness, and reproducibility of ML/AI models, as well as ethical concerns about equitable representation, training, and control. Our research project aims to demonstrate that KG and SV in combination are able to trace the reasoning paths of LLMs and ML models in literary (Hölderlin, Goethe, Celan), political science, and historical research contexts (climate change discourse). This improves AI-based research designs in the humanities to become more robust, reliable, and convincing after contentious recent debates (2019). The combination of KG and SV technology allows to critically “read”, compare, and measure the reasoning paths taken by an AI through knowledge nodes and feature weights involved in the predictions, revealing underlying methodologies, heuristics, biases embedded in an AI model, its training data, and pathways in human-readable form. Along with two exemplary applied case studies on literary corpora and historical sources using KG and SV, this project will contextualize this

undertaking with notes on the technological concept history of these XAI methods and their premises, thus making XAI accessible for humanists (e.g. game theory, semantic network graphs).

Title: “Agent-Driven, Game-Based Learning: Personalized CS Education for Students with Reading Disabilities.”

Target: National Science Foundation (NSF), Discovery Research PreK-12 (DRK-12)

Size: 4 years, \$3mil

PIs: Corey Clark (Computer Science and Guildhall), Leanne Ketterlin Geller (Education), Steph Buongiorno (Guildhall), Yanjun Pan (Education), and Eric Larson (Computer Science)

Summary:

The need for student engagement and skill development in STEM concepts has never been greater. However, the lack of universal access support to students with reading disabilities can hinder individual progress towards obtaining these skills. This challenge was made even more difficult by the COVID pandemic, which affected the performance of students with reading disabilities and left instructors with the challenge of managing a wider range of student skills. When learning materials are appropriately accessible, students are able to meaningfully demonstrate their understanding in ways that are comparable to their peers. AI agents, powered by large language models (LLMs), offer a unique solution to providing universal access to students with reading disabilities. LLMs are highly flexible and adaptable, with an ability to autonomously process information and tailor learning experiences to individual student needs. This personalization enhances learning without compromising the generalizability of the AI agents’ design in which a singular agent design can be deployed and autonomously adapted to diverse scenarios.

Our project aims to extend the state-of-the-art in AI agent design and deploy AI agents in K-12 classrooms to promote equity and better prepare students for careers in a technology-driven world. This project builds on prior NSF-funded research, including the "Factory Craft" initiative in Minecraft, aimed at making STEM education accessible to diverse student populations. We propose to extend this research by designing and deploying two types of AI agents in K-12 classrooms: “Personalized Education Agents” and “Teacher Support Agents.” Integrated into game-based platforms like Minecraft, Personalized Education Agents will autonomously adjust learning experiences based on individual literacy needs, offering feedback and creating personalized game scenarios that align with each student’s abilities. “Teacher Support Agents” will act as personalized assistants, providing educators with clear insights into student progress through easy-to-read summaries and visualizations. Designed this way, we aim to bridge the gap between accessibility and effective STEM education.

Invited Talks and Lectures

2027	(Upcoming) “AI and the Historian’s Craft: Promise, Peril, and Practice in Research and Teaching,” Panel at American Historical Association (AHA), New Orleans. Panel with: Jo Guldi, Kalani Craig, Kera Lovell, Ian Milligan.
2026	(Upcoming) “How Global is AI’s Telling of Global History? The Opportunities and Risks in Summarizing Wikipedia.” Text Mining Across Space, Time, and Languages. Social Science History Association (SSHA). Atlanta, Georgia. Panel with: Jo Guldi and Jim Clifford.
2026	(Declined) Wikimania, Paris, France.
2026	“Who is the Working Class Socialist James Bond? Exploring Propaganda through Knowledge Graphs and AI-assisted ‘Forensic Reading’ in Bulgarian spy fiction,” <i>The Convergence of Artificial Intelligence, Cyber, and Intelligence on Global Security Interests</i> . International Association for Intelligence Education (IAFIE), Annual Global Conference, University of North Georgia. Panel with Adam Hanzel and Kiril Avramov.
2026	“Visualizing Consensus and Divergence in Wikipedia’s Telling of Global History,” Roundtable on Text Mining at <i>Southern Conference on Slavic Studies: Russia and Its Neighbors</i> , Georgetown University, Washington, D.C. Panel with: Ernest Wyciszkiewicz and Bartek Gajos.
2026	Humanities Advocacy Day, National Humanities Alliance (NHA), Washington D.C. at the House and Senate Offices on Federal Funding in Support of the Humanities
2025	“Modeling Worldviews: How Do Historian’s Tools Help Us Understand AI Bias,” Women in Technology (WIT), Emory University.
2025	“Taking Stock and Setting Agendas: A Workshop on Data, Digital Humanities, and Early American Studies,” Round Table on AI, Iona University and the Institute for Thomas Paine Studies
2025	“Introducing Democracy Viewer,” Invited Panelist at the Archives as Data Conference, Columbia University, hosted by Columbia’s History Lab and Columbia Libraries
2024	“Introducing Democracy Viewer,” DHLab, The University of Texas at Austin
2024	“Combating Human Trafficking with Technology,” World Affair Council of DFW, hosting U.S. State Department international visitors from India
2024	“Introducing the Democracy Viewer Web App,” History of Parliament Trust
2024	“Navigating Privacy, Storage, and Analysis in a Generative AI Enabled World,”

SMU Human Trafficking Data Conference, Sponsored by Thomson Reuters.

- 2024 Generative AI and Agents Integration into Human Trafficking Data and Analysis,” SMU Human Trafficking Data Conference, Sponsored by Thomson Reuters.
- 2024 “Rhetorical Tools for Storytelling,” Guildhall, Southern Methodist University
- 2023 “What is Machine Learning?,” Department of Civil and Environmental Engineering, Southern Methodist University
- 2022 “Political Representation: Tensions between Parliament and the People from the Age of Revolutions to the 21st Century,” Department of History, Academy of Finland
- 2022 “Parliamentary data and search interfaces,” Department of History, Academy of Finland
- 2022 “Data Storage and Data Processing,” Public Health, UT Southwestern
- 2022 “Text Mining and Data Analytic Techniques,” Public Health, UT Southwestern
- 2021 “Extracting Gendered Constructions with spaCy,” History, Southern Methodist University
- 2021 “Natural Language Processing and Understanding Metadata,” History, Southern Methodist University
- 2020 “An Introduction to Technical Writing,” Ethical Issues in Computing, Department of Computer Science, Southern Methodist University
- 2018 “Analyzing Gender and Agency using Grammatical Triples Extraction.” Think-Play-Hack, Southern Methodist University

Media Coverage

[“The I-Team: Video game helps researchers fight sex trafficking”](#), CBS Texas, 2023 **[TV]**

[“Researchers to present new tool for enhancing AI transparency and accuracy at conference”](#). Microsoft Network (MSN) **[Online Article]**

[“SMU creating human trafficking data ‘warehouse’”](#), ScienceX, 2023 **[Online Article]**

[“‘Dark Shadows’ Video Game Lights up Human Trafficking Research at SMU”](#), Dallas Innovates, 2023 **[Online Article]**

[“SMU researchers to present new tool for enhancing AI transparency and accuracy at IEEE Conference”](#), Newswise, 2023 **[Online Article]**

[“SMU researchers to present new tool for enhancing AI transparency and accuracy at IEEE Conference”](#), EurekAlert!AAAS, 2024 **[Online Article]**

[“SMU researchers to present new tool for enhancing AI transparency and accuracy at IEEE Conference”](#), ScienMag, 2024 **[Online Article]**

[“Researchers to present new tool for enhancing AI transparency and accuracy at Conference”](#), Lifeboat Foundation, 2024 **[Online Article]**

[“Researchers to present new tool for enhancing AI transparency and accuracy at Conference”](#), TechXplore, 2024 **[Online Article]**

[“Researchers to present new tool for enhancing AI transparency and accuracy at Conference”](#), AI Transformation Hub, 2024 **[Online Article]**

[“Researchers to present new tool for enhancing AI transparency and accuracy at Conference”](#), Apex Life Hub, 2024 **[Online Article]**

[“SMU researchers to present new tool for enhancing AI transparency and accuracy at IEEE Conference”](#), Bioengineer, 2024 **[Online Article]**

[“SMU researchers to present new tool for enhancing AI transparency and accuracy at IEEE Conference”](#), Milano All News, 2024 **[Online Article]**

[“Researchers Unveil GAME-KG Framework to Boost AI Transparency and Precision”](#), MultiPlatform AI, N.d. **[Online Article]**

[“Researchers to present new tool for enhancing AI transparency and accuracy at Conference”](#), Lifeboat Foundation Russia, 2024 **[Online Article]**

[“SMU Unveils AI Transparency Tool at IEEE Conference”](#), Mirage News, 2024 **[Online Article]**

[“Researchers to present new tool for enhancing AI transparency and accuracy at conference”](#), Tech Post and Science, 2024 **[Online Article]**

[“University Creating Human Trafficking Data Warehouse Using Video Game Tech”](#), Mirage, 2024 **[Online Article]**

[“https://www.dallaschamber.org/blog/smu-team-creating-a-data-warehouse-to-fight-human-trafficking/”](https://www.dallaschamber.org/blog/smu-team-creating-a-data-warehouse-to-fight-human-trafficking/), Dallas Regional Chamber, 2025 **[Online Article]**

[“https://www.promptlayer.com/research-papers/a-framework-for-leveraging-human-computation-gaming-to-enhance-knowledge-graphs-for-accuracy-critical-generative-ai-applications”](https://www.promptlayer.com/research-papers/a-framework-for-leveraging-human-computation-gaming-to-enhance-knowledge-graphs-for-accuracy-critical-generative-ai-applications), PromptLayer, 2024 **[Online Article]**

<https://www.balancedmediatechnology.com/collaborative-initiative-aids-in-human-trafficking-research/>”,
BalancedMedia Technology, 2024 [**Online Article**]

<https://rviews.rstudio.com/2023/02/28/january-2023-top-40-new-cran-packages/>”, R Views, 2023 [**Online Article**]

<https://www.r-bloggers.com/2023/02/january-2023-top-40-new-cran-packages/>” Rbloggers, 2023 [**Online Article**]

Teaching

2026	QTM 340: Text as Data. Undergraduate Course. Data and Decision Sciences, Emory University. Supported Jo Guldi.
2026	ECON & HLTH 470: Research in Health Economics. Undergraduate Course. Department of Economics, Emory University. Co-taught with Ian McCarthy.
2025	DSCI 340: Text as Data. Undergraduate Course. Data and Decision Sciences, Emory University
2025	I 304 : Python Programming for Informatics. Undergraduate Course. iSchool, The University of Texas.
2024	HGME 6276: Thesis II: Level Design. Graduate Course. Guildhall, Southern Methodist University.
2024	HGME 6276: Thesis II: Art. Graduate Course. Guildhall, Southern Methodist University.
2024	HGME 6276: Thesis II: Software Development. Graduate Course. Guildhall, Southern Methodist University.
2024	HGME 6276: Thesis II: Production. Graduate Course. Guildhall, Southern Methodist University.
2024	HGME 6377: Thesis III: Software Development. Graduate Course. “gaia: A GEN-AI Gaming Framework for creating Dynamic Gameplay Content via Large Language Models.” Guildhall, Southern Methodist University.
2024	HGME 6377: Thesis III: Level Design. Graduate Course. “Unveiling New Realms: Enhancing Procedural Narrative Generation and NPC Personalization using AI.” Guildhall, Southern Methodist University.
2024	HGME 6377: Thesis III Production. Graduate Course. “Chinese PC Gamers’ Motivation, Game Preferences, and Publishing Advice for Non-Chinese Game Companies.” Guildhall, Southern Methodist University.
2024	CS 5352: Senior Design II. Undergraduate Course. Computer Science, Southern Methodist University. Topic: Taught a year-long digital humanities project with 3 enrolled undergraduates.
2023	Instructor, Guildhall Academy Video Game Design (high school students), Guildhall, Southern Methodist University.
2023	CS 5351: Senior Design I. Undergraduate Course. Computer Science, Southern Methodist

	University. Topic: Taught a year-long digital humanities project with 3 enrolled undergraduates.
2023	HGME 6276: Thesis II Software Development. Graduate Course. "gaia: A GEN-AI Gaming Framework for creating Dynamic Gameplay Content via Large Language Models." Guildhall, Southern Methodist University.
2023	HGME 6276: Thesis II Level Design. Graduate Course. "Unveiling New Realms: Enhancing Procedural Narrative Generation and NPC Personalization using AI." Guildhall, Southern Methodist University.
2023	HGME 6592: Team Game Production I. Graduate Course. Topic: Replicability and Version Control. Guildhall, Southern Methodist University.
2023	Teaching Fellow, Foundations and Applications of Humanities Analytics, Santa Fe Institute.
2023	HGME 6268: Directed Studies in Creative Narrative Writing for Video Games. Graduate Course. Guildhall, Southern Methodist University.
2022	HGME 6266: Directed Studies in Creative Narrative Writing for Video Games. Graduate Course. Guildhall, Southern Methodist University.
2022	HIST 3380: Topics in Digital History, Department of History, Southern Methodist University.
2022	Instructor, Guildhall Academy, Southern Methodist University.
2022	Teaching Fellow, Foundations and Applications of Humanities Analytics, Santa Fe Institute.
2021	HIST 3380: Topics in Digital History, Department of History, Southern Methodist University.
2018	WRTR 1312: Introduction to Academic Writing, Department of English, Southern Methodist University. (1 section)
2017	WRTR 1311: Foundations of Writing, Department of English, Southern Methodist University.
2017	PRW 2144: SCUBA, Department of Physical Fitness, Southern Methodist University.
2016-17	ENGL 102: Composition and Rhetoric. Department of English, West Virginia University (4 sections).
2015-16	ENGL 101: Introduction to Composition and Rhetoric. Department of English (4 sections).
2014	EXSA 1249: Advanced Scuba Diving. Department of Exercise and Sport Activity, The University of Texas at Arlington (2 sections).

2014 Scuba Diving. Texas Woman's University. Department of Fitness and Recreation.

2012-14 EXSA 1249: Scuba Diving. Department of Exercise and Sport Activity, The University of Texas at Arlington (6 sections).

Working Groups and Summer Schools

2026	(Declined) Summer Institute in Computational Social Science (SICSS), University of Pennsylvania, Selected Participant.
2024-Present	DH Research Group #1, AI Safety and Security in Digital Humanities (AIDHSec). Humanities Institute. The University of Texas at Austin.

Service

Committees

2026-present	AI and Computing Support Taskforce, Emory University
2025-2026	Committee Member, Faculty Computing Governance Committee, Emory University
2019	Committee Organizer, Interdisciplinary Nineteenth-Century Studies Conference, Southern Methodist University

Organizations

2024- present	High School Program on Democracy Viewer
---------------	---

Other Service

2026	Survey Participant, Interactive Fiction and Generative AI: Practitioner Interview, Cambridge MBA thesis, Suphil Park
2026	Peer Reviewer, <i>Historical Methods: A Journal of Quantitative and Interdisciplinary History</i>
2025	Peer Reviewer, <i>Historical Methods: A Journal of Quantitative and Interdisciplinary History</i>
2025	Peer Reviewer, <i>The Journal of Interactive Technology and Pedagogy</i>
2024	Track Co-Chair, SciPy, Social Science, Humanities, Economics, and Human Systems
2024	Peer Reviewer, IEEE Conference on Games, Long Papers
2024	Peer Reviewer, IEEE Conference on Games, Short Papers
2023	Track Co-Chair, SciPy, Computational Social Science and Digital Humanities